

Prathmesh Thakare

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Professional Summary

MSc AI graduate with 3 years of experience in production monitoring within the banking sector. Strong academic foundation in anomaly detection and image classification, supported by hands-on research projects using deep learning and statistical methods in Python. Experienced in root cause analysis, performance analytics, and collaborating with cross-functional teams to support business-critical systems. Seeking an opportunity to apply research and analytical skills to real-world ML problems.

Skills

- **AI & Machine Learning:** Anomaly Detection, Time-Series Analysis, Autoencoders, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, MLOps fundamentals
- **Programming:** Python, JavaScript, MySQL
- **Monitoring & Testing:** Dynatrace (APM, OneAgent, Root Cause Analysis), log Analysis, Incident Management
- **Tools:** JIRA, ServiceNow, BMC Remedy
- **Collaboration & Soft Skills:** Cross-functional Collaboration, Client Communication, Mentoring, Analytical Thinking, Problem Solving.

Work Experience

Quality Kiosk Technologies Pvt Ltd | Software Engineer

12/2021 – 08/2023

- Worked on Application Performance Management (APM) using Dynatrace Managed for Axis Bank and IndusInd Bank.
- Reduced transaction failures by 30% through proactive monitoring of applications such as Net Banking, Mobile Banking, IMPS/UPI, Savings/FD, and Core Banking.
- Performed Root Cause Analysis (RCA) for live production issues, collaborating with application and synthetic monitoring teams to minimise user impact.
- Delivered performance insights and dashboards to senior technology and business stakeholders to support strategic decision-making.
- Recognised with the "Stellar Award" for identifying and resolving a critical issue in a high-pressure environment.

Ruparel Realty India Pvt Ltd | Associate – IT

11/2020 – 12/2021

- Handled BMC Remedy tickets for issue resolution and request tracking, ensuring timely closure and SLA compliance.
- Managed incidents and RCA reporting, and coordinated weekly meetings with Service Delivery Management teams.
- Documented issue resolutions and streamlined support workflows, improving internal team efficiency.

Projects:

Time Series Anomaly Detection

03/2026 – Present

- **Problem Statement:** To study time-series anomaly detection techniques for identifying abnormal patterns in sequential data, such as network traffic, building on prior research experience in anomaly detection.
- **Techniques Used:** Researching statistical and deep learning approaches, including moving averages, ARIMA, LSTM-based autoencoders, and reconstruction-error methods, with a focus on data preprocessing, windowing, and model evaluation for sequential anomaly detection.

IndoFashion Image Classification

08/2025 – 12/2025

- **Problem Statement:** Developed a multi-class image classification solution from the IndoFashion dataset, addressing challenges of limited per-class training data and class similarity.
- **Techniques Used:** Implemented a transfer learning approach using TensorFlow with MobileNetV2, combined with stratified sampling, custom data loading, augmentation, and a two-phase training strategy (feature extraction + fine-tuning).
- **Results Achieved:** Achieved ~78% test accuracy, outperforming ResNet50 (61%), with faster training time and better generalisation across most classes.

Anomaly Detection using GANs

04/2024 – 10/2024

- **Problem Statement:** Evaluated the effectiveness of GANs for detecting anomalies in network traffic data, on the CICIDS 2017 dataset (~3M network flow records) using PyTorch.
- **Techniques Used:** Trained a GAN model with data preprocessing, class balancing, and dimensionality reduction using PCA. Applied stratified sampling and cross-validation for model evaluation, and benchmarked its performance against a CNN model for comparison.
- **Results Achieved:** GANs faced stability and convergence challenges on highly imbalanced and high-dimensional data, with limitations related to training time and model collapse compared to CNNs, highlighting the importance of selecting model architectures based on data characteristics and practical deployment constraints.

Education:

Aston University, Birmingham (2025)

M.Sc. Artificial Intelligence with Business Strategy

Key Modules: Machine Learning, Deep Learning, Mathematics for AI, Leadership and Decision Making, Strategic Management.

Mumbai University, Mumbai (2020)

B.Sc. Information Technology

Programming Language – C, C++, Java, JavaScript, Python, DBMS, MySQL, Linux.